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A novel optimal wavelet filter banks for automated diagnosis of Alzheimer's disease and mild cognitive impairment using Electroencephalogram signals

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ABSTRACT

Electroencephalogram (EEG) of Alzheimer's disease (AD) patients show a slowing effect and less synchronization. EEG signal's transient and abrupt nature is captured from various mother wavelets. However, better performance can be obtained by balancing time–frequency localization in wavelet filters. We propose a new approach for designing filter banks based on optimal time–frequency localization of a four-step lifting structure (FSLS). First, we design a FSLS using Euler's Frobenius half-band polynomial (EFHBP). The perfect reconstruction condition and vanishing moments are imposed on EFHBP to achieve maximum flat half-band filters (HBFs). HBFs are optimized using a balanced uncertainty metric to obtain a time–frequency spread balance. Afterward, these HBFs are used in the FSLS to obtain the synthesis and analysis filter banks. The proposed time–frequency optimized biorthogonal wavelet filter banks (TFOBWFBs) achieved a balance between time–frequency localization. Further, these TFOBWFBs are applied to decompose the EEG signals of AD patients. Twenty different features were extracted from decomposed EEG subbands, of which twelve significant features were selected using Kruskal Walli's test. The machine learning models were trained and tested using obtained features with 10-fold cross-validation and leave-one-subject-out cross-validation. To validate this study, the proposed TFOBWFBs have been applied to two publicly available EEG datasets of mild cognitive impairment (MCI), AD, and healthy control (HC) subjects. The proposed TFOBWFBs achieved 98.90% accuracy for 2-way (AD vs. HC) and 96.50% for 3-way (AD vs. MCI vs. HC) classification using a support vector machine model with a 10-fold cross-validation. The proposed method outperforms the existing AD and MCI detection techniques. The performance of the proposed machine learning framework with optimization of wavelet filter banks is more significant and fast compared to previous studies. Also, the proposed framework can be applied to detect other neurodegenerative disorders.

1. Introduction

Alzheimer's disease (AD) is a widely detected neurodegenerative type of dementia. It affects the population which is above 65 years of age. Its common symptoms are memory loss and difficulties in thinking and performing daily activities [1]. AD is caused by the spread of non-solvable parts of the protein named plagues and tangles. AD became the seventh leading death disease worldwide in 2021 [2]. Almost 60 million people were living with AD in 2020. It is estimated to rise to 150 million in 2050 [3]. There is no cure for AD to date. However, AD progression can be slowed by treating the patient at an early stage. There is an intermediate stage between AD and a healthy controlled (HC) state commonly known as mild cognitive impairment (MCI). The MCI displays numerous symptoms similar to those seen in AD. However, it does not meet the complete diagnostic criteria for AD [4]. Though 40% of the world's population suffers from MCI, only 12% of it converted to severe AD. Therefore, it is necessary to have an easy, fast, less expensive, and robust automatic AD detection system at an early stage. A positive AD diagnostic provides time for the patient and his family to inform themselves about the disease, make financial decisions about the disease, and plan for future care and needs related to the patient.

Various traditional methods exist to make decisions on AD diagnosis, such as mini-mental state examination (MMSE), neuro-imaging, and clinical tests. However, these tests are time-consuming, complex, and subjective. Many researchers and clinicians have been looking towards Electroencephalogram (EEG) as a reliable tool to detect Schizophrenia, Parkinson's disease, stress levels [5], drowsiness, emotion [6], and

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epileptic seizures in the past decade. EEG is a low-cost, portable, non-invasive technique with good temporal resolution. EEG captures the brain's electrical variation using different electrodes placed on the head. The EEG has the most important information in the low-frequency bands, such as 0–4 Hz ( ), 4–8 Hz ( ), 8–12 Hz ( ), 12–16 Hz ($_1$), 16–32 Hz ($_2$), and 32–48 Hz ( ) [7]. AD shows major EEG effects, including complexity reduction, slowing, and perturbations in EEG synchrony. AD EEG is characterized by a decrease in the and and an increase in power of and bands [8,9].

The non-linear methods, including sample entropy [10], approximate entropy (ApEn), auto mutual information (AMI) [11], detrended fluctuation analysis (DFA) [12], and Lampel-Ziv complexity (LZC) [13], detrended moving average (DMA), distance-based LZC (dLZC) [14], generalized multiscale entropy (GMSE), multivariate multiscale entropy [15], Kolmogorov entropy [16], fuzzy entropy [17], phase synchrony, cross-correlation [18], periodic activity [19], and Hjorth parameters, have been reported to detect AD from EEG signals. However, these non-linear methods are parametric, and their performance is inconsistent in AD detection. Hong et al. [20] analyzed EEG power spectrum-based image features and brain networks of AD EEG signals. This method obtained 88.50% classification accuracy from the ensemble model [20]. However, this method is more complex and provides low performance. Pirrone et al. [21] developed a decision-making model for AD diagnosis based on filtering using finite impulse response. Their models filter the EEG signals with a cut-off frequency of 7 Hz and 16 Hz. The power difference of these frequency bands is used as features with support vector machine (SVM) for 3-way classification and obtained 86% accuracy. However, these results are not consistent with the time. Similarly, Swarnalatha [22] has investigated the hybrid greedy sandpiper-based deep learning model to detect AD and its stages, such as mild, moderate, and severe. It obtained the highest classification rate of 99.8%. However, this model is not applicable to other publicly available datasets [22]. These deep-learning models take more time to train and are computationally complex.

Different notions in the literature have been reported on time-frequency analysis to detect AD and MCI [23–29]. Fiscon et al. [23] employed the fast Fourier transform (FFT) and discrete wavelet transform (DWT) to extract the features from MCI, AD, and HC EEG signals. It has been reported that DWT provided the 92% of AD detection accuracy with a decision tree (DT) classifier. However, the 3-way classification performance was not reported [23]. Sharma et al. [24] have investigated different EEG biomarkers and wavelet-based power spectral density (PSD) to detect AD. However, they obtained only 89.9% classification accuracy. In another study [25], the spectral and complexity-based features have been extracted for AD detection. It is found that the K-nearest neighbor (KNN) has achieved 94% of accuracy. Similarly, FFT-based bump modeling and DWT-based features with SVM have been reported in [26]. They obtained 94% and 92% of classification accuracies using SVM and KNN, respectively. However, the hyper-parameters of machine learning models have not been tuned. Sanchez et al. [27] have calculated empirical wavelet transform (EWT)-based on three fractal dimensions features to detect AD and MCI. They reported a 90.3% classification accuracy using enhanced probabilistic neural networks. An FFT and DWT-based energy feature have been introduced to detect AD and achieved 97% accuracy using KNN [28].

Another study by Alsharabi et al. [30] developed a computer-aided diagnostic model to detect AD and its stages. The DWT-based statistical features were investigated from all classes of EEG signals and reported that their system is an ideal system [30]. However, the dataset used in [30] is relatively small. Hence, the trained machine learning models may be over-fitted. Fouad et al. [31] presented a DWT-based AD detection model. They employed 13 different features to obtain a good accuracy of 96.55% using the Naive Bayes algorithm. Moreover, the performance of ResNet-50 is also investigated and achieved 97.83% accuracy. However, the hyper-parameters of classification models have not been tuned [31]. Recently, an adaptive-flexible analytic wavelet transform (AFAWT) has been proposed by Khare and his group [32] for the automatic detection of AD. They utilized root mean square error to obtain the optimal parameters of AFAWT. Viccietti et al. [33] suggested the computer-aided model for the diagnosis of AD and MCI. They investigated fractal dimension, wavelet coherence, wavelet energy, quadratic entropy, quantile graphs, and visibility graphs from AD and probable AD EEG signals. This approach obtained 100% performance metrics. However, it is worth mentioning that the subjects under EEG recording were not submitted to a definitive pathological diagnosis of AD or health controls. Similarly, Safi et al. [29] explored the performance of Hjorth parameters with DWT and empirical mode decomposition (EMD) to detect AD. The Hjorth parameters features have been employed in linear discriminant analysis (LDA) and KNN. The KNN obtained an accuracy of 97.64% from 10-fold cross-validation and the lowest accuracy of 86% from leave-one-subject-out (LOSO) cross-validation [29]. Toural et al. [34] has used wavelet entropy and band power features to detect AD. They achieved a 94.44% accuracy. Sharma et al. [35] used iterative filtering to decompose AD EEG signals. The PSD, fractal dimension, and entropy features were captured from decomposed subbands. Different machine learning models have been trained and tested. The KNN obtained 92% accuracy. However, this method does not provide consistent results. Ding et al. [36] used complexity and spectral features for AD detection. They reported an 80.08% classification rate. Recently, Araujo et al. [37] performed wavelet packet decomposition to extract features with neural networks and obtained 82.2%. Trambaiolli et al. [38] used different spectral features with SVM and enhanced the accuracy up to 91.18% for AD detection. Ieracitano et al. [39] investigated continuous wavelet transform and bi-spectrum-based features to identify MCI and AD. They utilized various machine learning models, out of which the multi-layer perceptron classifier outperforms other models with 96.5%. Similarly, Miltiadous et al. [40] mentioned PSD-based features to detect dementia and obtained a 77.01% detection rate using random forest. To investigate the brain activity complexity in AD, a multivariate multiscale weighted permutation entropy was used in [41]. The 96.70% accuracy was obtained with a one-way analysis of variance test. A systematic literature review on the detection of AD from EEG signals has been presented in [42].

Most of the above-discussed methods used wavelets for the time–frequency analysis of EEG signals. Since EEG exhibits transient events, there is a need to have a tool that possesses good time–frequency localization. The Daubechies wavelets are commonly observed in various state-of-the-art techniques. However, it does not have arbitrary localized wavelet bases in frequency and time simultaneously. Therefore, various researchers have designed wavelet filter banks with optimal balance between time–frequency localization that performed better on EEG signal analysis. However, these filter banks have ripples in the wavelet and scaling functions. Moreover, there is no degree of freedom while designing the filters. Further, halfband filters (HBFs) can be designed using two-step, three-step, or four-step filter banks. However, a four-step lifting structure (FSLS) has overcome the limitations of two-step and three-step HBFs. Therefore, we designed a novel class of time–frequency optimized biorthogonal wavelet filter banks (TFOBWFBs) with a new Euler–Frobenius halfband polynomial (EFHBP). The proposed TFOBWFBs are further utilized to detect AD and MCI from HC EEG signals. This presented work supports physicians, neurologists, and mental health counselors in their decision-making on AD and its stages. Below are the key contributions of the current study,

1. Designing a new class of filter banks using Euler–Frobenius halfband polynomial in the lifting structure to obtain perfect reconstruction.
2. Designing a new class of filter banks with optimal balance between time-frequency localization using a four-step lifting structure and balanced uncertainty metric.
3. Investigating the performance of TFOBWFBs and machine learning algorithms by producing a model for the detection of AD and MCI.

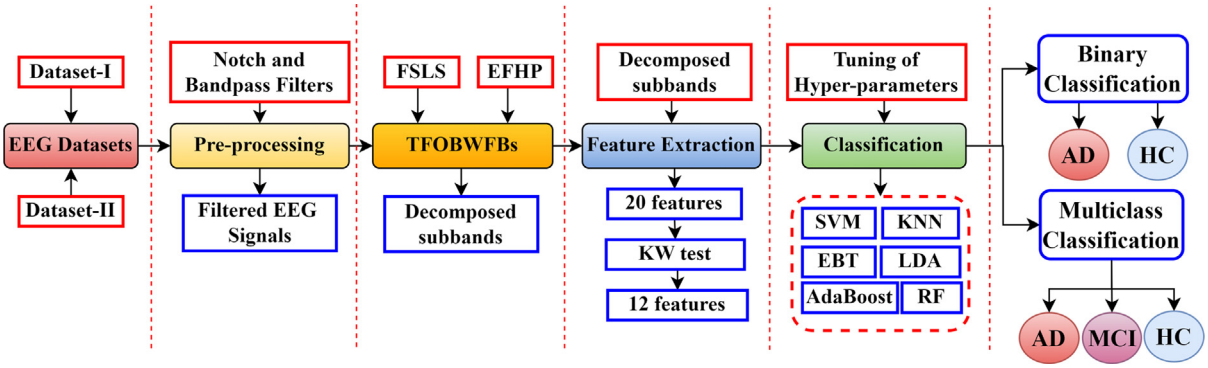

Fig. 1. Flow of the proposed method to detect the AD and MCI.

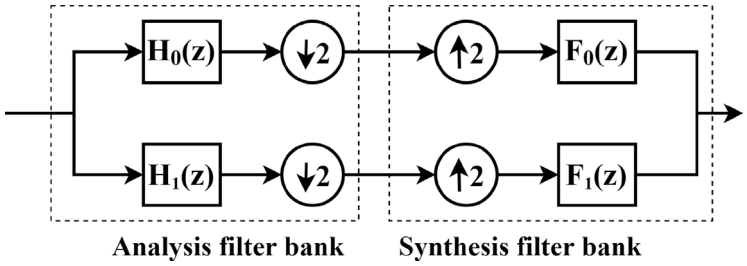

Fig. 2. Structure of 2-channel filter bank.

4. Estimating the significant features through desirable statistical methods.
5. Assessing the performance for various machine learning models with tuned hyper-parameters to detect AD and MCI.
6. Enhancing the performance of AD detection compared to other existing methods.

The following sections of this paper are structured as follows: The details of the proposed new class of TFOBWFBs and automatic AD detection techniques are presented in Section 2. The performance of the proposed study on the detection of AD has been provided in Section 3. The comparative analysis and features of the proposed method have been highlighted in the discussion Section 4.

2. Methodology

Fig. 1 illustrates the proposed technique. Initially, the artifacts in EEG signals were removed using an FIR filter with 0.5 Hz and 100 Hz cut-off frequencies. A 60 Hz notch filter was utilized to eliminate the power-line frequency. Further, all EEG recordings have been decomposed by employing the proposed novel TFOBWFBs into six subbands, namely , , , $_1$, $_2$, and . Afterward, twenty different features were extracted from each SB for AD classification. Next, the importance of these features has been checked by the statistical Kruskal Walli's test. Finally, various machine learning models, namely KNN, random forest (RF), AdaBoost, LDA, ensemble bagged tree (EBT), and SVM have been trained and tested with LOSO and the 10-fold cross-validation technique.

2.1. Filter bank fundamentals

The simple structure of a two-channel biorthogonal wavelet filter bank (BWFB) is formed with four filters, namely, analysis lowpass ($H_0(z)$), synthesis lowpass ($F_0(z)$), analysis highpass ($H_1(z)$), and synthesis highpass ($F_1(z)$) filters. The two-channel filter bank (TCFB) is shown in Fig. 2. The TCFBs are known as perfect reconstruction filter banks if they follow the given conditions:

$$H_1(z)F_1(z) + H_0(z)F_0(z) = cz^{-l} \tag{1}$$

$$F_1(z)H_1(-z) + F_0(z)H_0(-z) = 0 \tag{2}$$

where l is the amount of delay and c is constant. The product filter $P(z) = F_0(z)H_0(z)$ follows the special class of filters called halfband filters. To achieve the perfect reconstruction condition, all four filters should hold the following relation which is given as:

$$F_0(z) = zH_1(z) \quad \text{and} \quad F_1(z) = zH_0(-z) \tag{3}$$

The wavelet ((t)) and scaling ((t)) functions of analysis filter banks are provided by the following relations:

$$(t) = \frac{2}{F_0(!)\delta_{!=0}} \sum_n h_1(n) \ (-n + 2t) \tag{4}$$

$$(t) = \frac{2}{H_0(!)\delta_{!=0}} \sum_n h_0(n) \ (-n + 2t) \tag{5}$$

where $h_1(n)$ and $h_0(n)$ are impulse responses of analysis highpass (HPF) and lowpass (LPF) filters, respectively.

2.2. Proposed design of TFOBWFBs

A novel class of 2-channel BWFBs has been proposed in [43,44], which is known as Four-Step Lifting Structure, as depicted in Fig. 3. The two LPFs of analysis and synthesis structure for ($L = 4$) are expressed as:

$$H_0(z) = C_0 \ 1 + (p_3 + p_0)T(z) + p_3p_1T^2(z) + p_3p_2p_1T^3(z) \tag{6}$$

$$F_0(z) = C_1 \ 1 + (p_4 + p_2)T(z) + (p_2p_1 + p_4p_1 + p_4p_3)T^2(z) + p_4p_1p_3T^3(z) + p_2p_1p_4p_3T^4(z) \tag{7}$$

where C_0 and C_1 are used to take care of dc and ac filter gains, p; p_1; p_2; p_3, and p_4 are lifting parameters. These parameters are evaluated by imposing the constraints at different frequencies of the filter. $H_0(-1) = H_1(1) = 0$ ensures no dc leakage. Hence, the lifting parameters are given as follows:

$$p_3 = \frac{1 - p_1}{1 - p_2 + p_1p_2}; \quad p_4 = \frac{1}{2}(-1 - 2p_1p_2 + p_2^2 - p_1^2p_2^2) \tag{8}$$

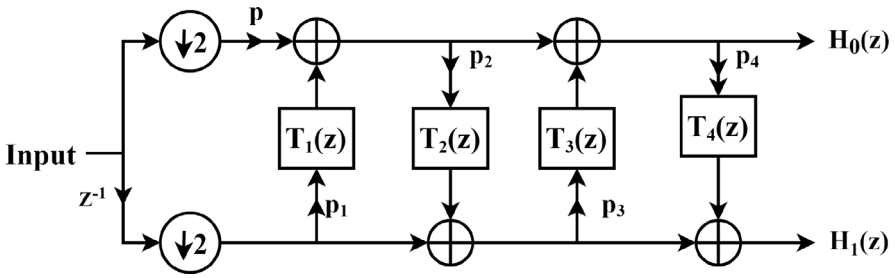

Fig. 3. 4-step lifting scheme for analysis FB.

by imposing $H_0(1) = H_1(1) = 0$, the normalization constants C_0 and C_1 are given by,

$$C_0 = \frac{1}{2}G_{dc}(1 - p_2 + p_1p_2); \quad C_1 = \frac{G_{dc}}{(1 - p_2 + p_1p_2)} \tag{9}$$

where G_{dc} is desired gain. It is $G_{dc} \approx 1$ in most of the applications. To maintain the biorthogonality, the p_2 in terms of free parameter p_1 is: $\frac{1}{p_1 - 1}$. The lifting filters as shown in Fig. 3 are considered identical i.e. $T_4(z) = T_3(z) = T_2(z) = T_1(z) = T(z)$. The identical lifting filter $T(z)$ is to be designed using EFHBP. It is defined as [45]:

$$T(z) = \sum_{k=0}^{M} \varphi(k+1)z^{-k} \tag{10}$$

Polynomial order M can be computed from [45]:

$$\varphi(k+1) = \sum_{m=0}^{k} (-1)^m \binom{M+2}{m} (k+1-m)^{M+1} \tag{11}$$

To get the scaling function smoothness, vanishing moments N and one degree of freedom α_k are added to modify the EFHP. It is given as [46]:

$$P(z) = (1 + z^{-1})^N T(z) \sum_{k=0}^{D} \alpha_k z^{-k} \tag{12}$$

Where, $K = M + D + N$ is the $P(z)$ order and $D = K/2 - 1$. The α_k is computed using the perfect reconstruction condition on (12). Then, the TFOHBFB is given by:

$$Q(z) = \sum_{k}^{L} \alpha_k R_k(z)^M + P(z) \tag{13}$$

In order to evaluate the second degree of freedom α_k, an interpolating polynomial R_k as defined in [45] is used, with K being selected as $K = D/2$ in our case. To balance time localization (t^2) and frequency localization (ω^2), the EFHBP is optimized by utilizing the balanced uncertainty metric. The balanced uncertainty metric is defined as [45]:

$$\mathfrak{B} = \omega^2 + a^2 t^2 \tag{14}$$

where t^2 and ω^2 are defined as follows [47]:

$$t^2 = \frac{1}{\sum_{n=0}^{K} q^2(n)} \sum_{n=0}^{K} n^2 q^2(n) \tag{15}$$

$$\omega^2 = \frac{2}{3} + 4 \sum_{n=0}^{K-1} \sum_{m=n+1}^{K} \frac{(-1)^{m-n}}{U(m-n)^2} q(m)q(n) \tag{16}$$

where, impulse response $q(n)$ is obtained from EFHBP. By puting (15) and (16) in (14), we obtain a ratio of quadratic equations:

$$\mathfrak{B} = \frac{^2+12 \sum_{m}^{3} \sum_{n}^{3} (-1)^{m-n}/(m-n)^2 q(m)q^2(n) + 3a^2 \sum_{n}^{3} q^2(n) + \sum_{n=0}^{3K} q^2(n)}{3 \sum_{n}^{3} q^2(n)} \tag{17}$$

To minimize the objective function of balanced uncertainty metric (17), we take first-order derivative of B with respect to optimized value $\alpha_{D/2}$ and equate it to null value, as follows:

$$\frac{\partial \mathfrak{B}}{\partial} = \frac{Y_{nr}Y_{dr} - Y_{dr}Y_{nr}}{(Y_{dr})^2} = 0 \tag{18}$$

where, Y_{dr} and Y_{nr} are partial derivatives of numerator Y_{nr} and denominator Y_{dr} of (17). Hence, the minimization of (18) produces a set of linear equations:

$$Y_{nr}Y_{dr} - Y_{dr}Y_{nr} = 0 \tag{19}$$

The optimized value of can be obtained by solving Eq. (19) iteratively using Algorithm 1. If we substitute the optimized value of into Eq. (13), we get the filter coefficients of optimized TFOHBF $Q(z)$. Then, the coefficients of optimized filters are designed using identical kernels of an FSLS as $T_0(z) = T_1(z) = T_2(z) = z^{-K/2}Q(z) - 1$. The design process of TFOBWFBs is illustrated in the given design example.

Algorithm 1 Algorithm for optimizing by solving the objective function.

Require: α_k, a^2.
1: Initialize $\alpha_i = 0$.
2: Compute: Y_{nr} and Y_{dr}.
3: Compute: Y_{nr} and Y_{dr}.
4: Solve (19) for α_{i+1}.
5: Update: $\alpha_i \leftarrow \alpha_{i+1}$.
6: Is $\alpha_i = \alpha_{i+1}$? Yes: Continue, No: Go to Step 2.
7: Compute: Filter vector **q**(n) using (13)

2.2.1. Design examples

To obtain the 6th order TFOBWFBs filters, the various parameters are selected as $N = 3$, $M = 1$, and $L = 2$. The lifting parameters are selected as: $p = 1$; $p_1 = 0{:}25$; $p_2 = -0{:}5333$; $p_3 = 0{:}5357$, and $p_4 = -0{:}2333$. The gains are $C_0 = 0{:}7000$ and $C_1 = 0{:}7143$. Then, $P(z) = (1+z^{-1})(z^{-3} + 11z^{-2} + 11z^{-1} + 1)(\alpha_0 + \alpha_1 z^{-1} + \alpha_2 z^{-2})$. By imposing the perfect reconstruction to above $P(z)$, the parameters are found as: $\alpha_0 = -1/238$; $\alpha_1 = 1/2$; $\alpha_2 = -1/238$. By putting $k = 1$ in (13), the $q(n)$ is obtained which is given by,

$$q(n) = \{(-0{:}0670 + 3\alpha_1)/32; 0; (8{:}0670 - 3\alpha_1)/32; 0{:}5\}:$$

By imposing the balanced uncertainty metric on $Q(z)$ and following algorithm 1 with $a^2 = 0{:}2$, the optimal value of α_1 is obtained as $-0{:}4$.

The frequency response of proposed 19/25 filters with optimal balance between time-frequency localization is plotted in Fig. 4. This frequency response is compared with the frequency response of existing 19/25 filter banks in [43,44]. It is noted that TFOBWFBs have achieved better frequency response than existing filter banks. The corresponding wavelet and scaling functions of the proposed 19/25 length filters are plotted in Figs. 5 and 6. It is clear that the existing filter banks have ripples in the scaling and wavelet functions that reduce the regularity of HBFs. However, the proposed 19/25 TFOBWFBs achieved smoothness and regularity in wavelet and scaling functions with no ripple. Hence, there will not be any loss in the EEG signal decomposition.

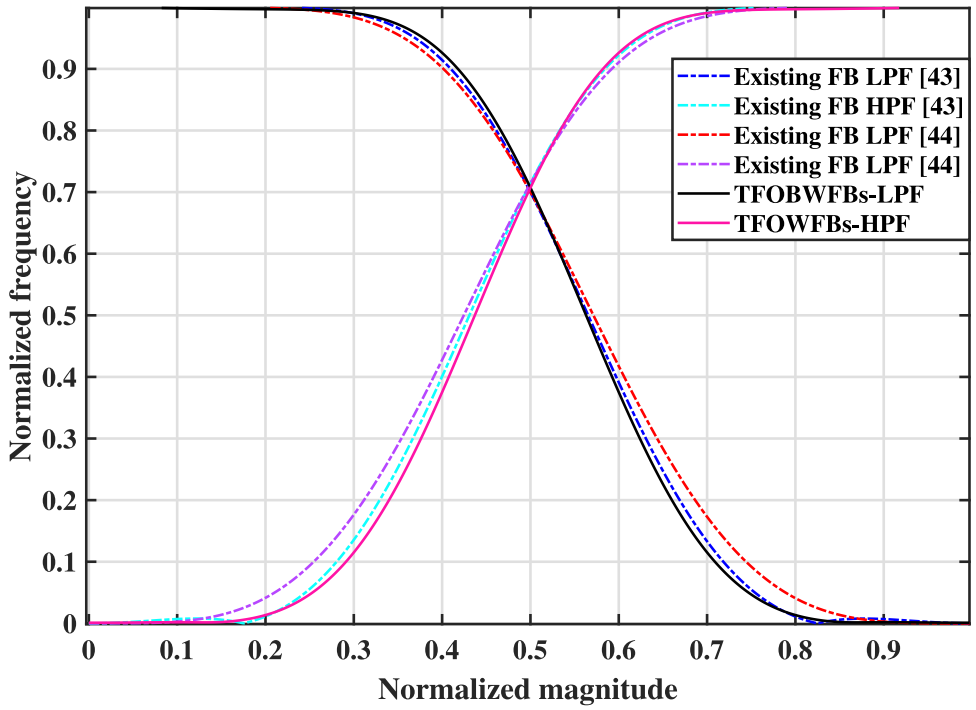

Fig. 4. Normalized frequency response comparison of designed 19/25 filters with existing filter banks.

Fig. 5. Wavelet function comparison of designed 19/25 filters with earlier 19/25 filter banks.

Fig. 6. Scaling function comparison of designed 19/25 filters with earlier 19/25 filter banks.